Polynomially many total mistakes for Bayesian neural-network learning

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Abstract

For uniformly sampled Boolean inputs, the strong-prior, above-mean-field, and mean-field learning results imply a stronger conclusion than eventual fixed accuracy: the total number of mistakes over an infinite learning stream has a polynomial bound, independent of learner width. The bound holds in expectation and with high probability, and applies both to posterior sampling and to counting a prediction as a failure unless the correct label has posterior probability greater than 2/3. The new statistical argument works for any consistency posterior whose relevant functions have a short finite description, even if their prior weights are extremely unequal. Deleting repeated inputs reduces the infinite iid stream to a uniformly random permutation. A harmonic bound over its random prefixes then gives an overhead of order the input dimension. We state precisely the structural facts imported from the neural-network papers, give explicit budgets, and distinguish this result from an arbitrary-order mistake guarantee. A prediction-preserving neuron-merging restriction also retains polynomial dependence on depth. These new cumulative conclusions have Lean proofs for the core statistical theorem and three boxed neural applications. Optional extensions have the narrower coverage recorded in the repository.

1 Prediction protocol and the result

Let $\mathcal{X}$ be a finite set of size $m \geq 2$, let $f : \mathcal{X} \to \{-1, 1\}$, and let π be a prior on Boolean functions with $\pi(f) > 0$. Before round j, the learner has seen the true labels of the preceding inputs and forms the exact consistency posterior Π_{j-1}. The next input X_j is then presented, a prediction is made, and its true label $f(X_j)$ is revealed. Every label is revealed, including after a correct prediction. Define

$$q_j = \Pi_{j-1}\{g : g(X_j) \neq f(X_j)\}, \qquad S = \sum_j q_j.$$

Thus S is cumulative posterior error probability. If the learner draws a function from the current posterior at each round, its total number of incorrect labels is denoted Z. The expected values of Z and S agree. For a fixed $0 < \theta < 1$, define the confidence-failure count

$$F_\theta = \sum_j \mathbf{1}\{\Pi_{j-1}(g(X_j) = f(X_j)) \leq \theta\}.$$

The $> 2/3$ confidence convention is $F_{2/3}$. Pointwise,

$$F_\theta \leq S/(1 - \theta). \tag{1}$$

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Any majority classifier's number of errors is at most $2S$, including arbitrary decisions at ties. These inequalities allow posterior mixtures; they require neither a unique minimum-complexity function nor posterior concentration on one function.

We consider either a uniformly random permutation of $\mathcal{X}$, used once, or an infinite iid stream uniform on $\mathcal{X}$. In the latter protocol every repeated input has posterior correctness probability one, and duplicating a hard conditioning constraint leaves the posterior unchanged. The order of first appearances is a uniformly random permutation. Consequently the total variables S, F_θ, Z have the same laws in the two protocols (using fresh posterior randomness on first appearances). We can therefore prove all bounds for m rounds without replacement.

For $\mathcal{X} = \{-1, 1\}^d$, the neural-network consequences are:

- At every fixed depth, the strong-prior learner has a total mistake bound polynomial in d and bounded-weight target-network complexity, at every width above that complexity, once its variance is sufficiently small at that fixed width.
- At every fixed depth and every fixed $\gamma > 1$, the specified finite-variance learner has the same type of polynomial bound, uniformly over all admissible widths. Its polynomial may depend on γ.
- The finite-variance mean-field tanh learner, $\gamma = 1$, also has the same polynomial bound at every admissible width.

High-probability bounds are polynomial in $\log(1/\delta)$ as well. The theorem does not replace uniform random input order by arbitrary order.

2 A cumulative bound from a finite relevant class

Write $\mathsf{H}_r = \sum_{k=1}^{r} 1/k$. The following theorem is the new statistical ingredient. In particular, it does not bound mistakes by the negative log prior probability of the target.

Theorem 1 (Random-order cumulative errors)**.** *Suppose A is a finite set of Boolean functions containing f and*

$$\log |A| \le h, \qquad \pi(A^c) \le \beta\pi(f), \qquad 0 \le \beta < 1. \tag{2}$$

Under either uniform input protocol above,

$$\mathbb{E}S = \mathbb{E}Z \le B_E := 1 + 2(h+1)\mathsf{H}_{m-1} + m\beta. \tag{3}$$

For $0 < \delta < 1$, put

$$a = h + \log(2m/\delta), \qquad B_\delta = 1 + 2a\mathsf{H}_{m-1} + m\beta, \qquad U_\delta = (e-1)B_\delta + \log(2/\delta).$$

Then $\mathbb{P}\{S > U_\delta\} \le \delta$ and $\mathbb{P}\{Z > U_\delta\} \le \delta$. These are separate probability statements; replace δ by $\delta/2$ to require both simultaneously. Equation (1) gives the corresponding bounds for every confidence threshold and for majority errors.

Proof. After any correctly labeled history, the posterior denominator is at least $\pi(f)$. Hence the posterior mass outside A is at most β, uniformly over all histories.

Let S_n now denote the set of the first n inputs of the permutation, and let

$$R_n = \max_{g \in A:\, g = f \text{ on } S_n} \frac{1}{m} |\{x : g(x) \ne f(x)\}|.$$

The maximum exists because f is included. A function with k errors survives a uniform size-n prefix with probability

$$\frac{\binom{m-k}{n}}{\binom{m}{n}} \le (1-k/m)^n \le e^{-nk/m},$$

with survival probability zero when $n > m-k$. A union bound gives

$$\mathbb{P}\{R_n > u\} \le \min\{1, e^{h-nu}\}, \qquad \mathbb{E}R_n \le \frac{h+1}{n} \quad (1 \le n < m). \tag{4}$$

All errors of a surviving function are among the remaining $m-n$ inputs. Conditional on the history, the expected posterior error on the next uniformly chosen remaining input is therefore at most

$$\frac{m}{m-n} R_n + \beta. \tag{5}$$

The first round costs at most one. Summing expectations and using

$$\sum_{n=1}^{m-1} \frac{m}{n(m-n)} = 2\mathsf{H}_{m-1}$$

proves (3). For posterior sampling, condition additionally on the current input before taking the expectation of the sampled label.

For high probability, let $E_n = \{R_n \le a/n\}$. By (4), the probability that any E_n fails is at most $(m-1)e^{h-a} \le \delta/2$. Stop counting at the first failed prefix event. More precisely, replace the round-$n+1$ variable Y_{n+1} by $Y'_{n+1} = Y_{n+1}\mathbf{1}\{E_1, \ldots, E_n\}$, with the indicator one at $n = 0$, where Y is either q or the sampled error indicator. The indicator is known before the next input. The sum of conditional expectations of Y' is at most B_δ, by (5). For any adapted variables $0 \le Y'_j \le 1$, the elementary bound

$$\mathbb{E}[e^{Y'_j} \mid \mathcal{F}_{j-1}] \le 1 + (e-1)\mathbb{E}[Y'_j \mid \mathcal{F}_{j-1}] \le \exp((e-1)\mathbb{E}[Y'_j \mid \mathcal{F}_{j-1}])$$

shows that $\exp(\sum_j Y'_j - (e-1)\sum_j \mathbb{E}[Y'_j \mid \mathcal{F}_{j-1}])$ is a nonnegative supermartingale. Markov's inequality bounds $\mathbb{P}\{\sum Y'_j > U_\delta\}$ by $\delta/2$. On all good prefixes, $\sum Y'_j = \sum Y_j$; a union bound proves the claim. □

For $m = 2^d$, take $\beta = 1/m$. Then $\mathsf{H}_{m-1} \le 1 + d\log 2$, so

$$B_E \le 2 + 2(h+1)(1 + d\log 2), \qquad U_\delta = O((h + d + \log(1/\delta))(d+1)). \tag{6}$$

The relevant description bound need only remain polynomial when $\log(1/\beta) = d\log 2$; it need not be independent of the desired tail.

What is uniform in width? The numerical bounds below hold for each admissible learner width with the same constants. The high-probability theorem is applied to a fixed learner and its posterior randomness. It does not assert that one event simultaneously controls independently sampled predictions of every learner.

3 The direct prior-mass bound

The smaller-width branches of the finite-variance neural-network proofs use a different, simpler estimate.

Lemma 2 (Exact Bayesian telescoping)**.** *For every input sequence, including adaptively chosen inputs,*

$$S \le \sum_j -\log(1-q_j) \le -\log \pi(f) =: J.$$

Thus $F_\theta \le J/(1-\theta)$ deterministically. Posterior sampling satisfies $\mathbb{E}Z \le J$, and

$$\mathbb{P}\{Z > (e-1)J + \log(1/\delta)\} \le \delta.$$

Proof. The product $\prod_j (1-q_j)$ is the prior probability that a function fits all revealed labels, and is at least $\pi(f)$. Use $q \le -\log(1-q)$. The sampled-error assertion follows from the same exponential supermartingale as above, conditional on each revealed query. Finite-horizon bounds extend to infinite streams by monotone convergence. □

This lemma alone generally costs a factor growing with neural-network width and inverse prior variance. The relevant-class theorem avoids that factor at large widths; the lemma covers the remaining widths where its cost is already polynomial.

4 Application to the three neural-network regimes

4.1 Model and imported structural results

The domain is $\{-1,1\}^d$. A depth-L network has $L-1$ hidden layers of width N, a constant input coordinate, bounded 1-Lipschitz hidden activation, and a linear readout. Define

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1}\|W^s\|_F^2 + N\|W^L\|_2^2.$$

The raw Gaussian is conditioned, before training, on $|z_w(x)| > 1/2$ for every input. It therefore induces a prior on Boolean sign functions. Training uses exact label agreement. Let $C = C_L(f)$ be the least width of a depth-L implementation with strict margin $1/2$ and all weights bounded in magnitude by that width.

We use the following proved structural facts from [1, 2, 3]. These are imports, not newly established Gaussian-compression theorems. Their explicit scalar budgets appear in Appendix A.

1. There is a common finite set containing every sign function whose minimum weighted energy is at most J, with log cardinality at most $F_L(d,J)$. This is polynomial in d, J at fixed L. Cloning gives $\kappa_N(f) \le B_L(d,C) := d(d+1)C^2 + LC^4$.

2. A raw target neighborhood has mass at least $\exp(-AN^\gamma/2)$ for every $N \ge C$, where $A = A_L(d,C) \ge 1$ is polynomial at fixed depth and the Gaussian variances are $N^{1-\gamma}/d, N^{-\gamma}, N^{-\gamma-1}$.

3. At fixed $\gamma > 1$, for $K \ge 1$ and widths satisfying

$$N^{\gamma-1} \ge 8(d+1), \qquad N^{\gamma-1} \ge 64 a_K^{-2}, \qquad a_K = [8L(\sqrt{K}+1)^{L-2}]^{-1},$$

the raw total mass of full-margin functions with minimum energy greater than $(L+3)K$ is at most $Le^{-KN^\gamma/8}$.

4. For tanh at $\gamma = 1$, if $K \geq 4$ and $N \geq T_L(d, K)$, the raw total mass of full-margin functions outside a common finite class of log cardinality at most $G_L(d, K)$ is at most $Le^{-NK/8}$. Both T_L and G_L are polynomial at fixed depth.

The last two bounds concern total exceptional mass, not one exceptional function at a time. This distinction is essential for this application.

4.2 Strong-prior limit

In the strong-prior parameterization, the energy coordinates are independent centered Gaussians of variance σ^2. At fixed width,

$$2\sigma^2 \log \mu_{N,\sigma}\{w : g(x)z_w(x) > 1/2 \ \forall x\} \longrightarrow -\kappa_N(g).$$

There are finitely many Boolean functions. Hence, with $A_{N,f} = \{g : \kappa_N(g) \leq \kappa_N(f)\}$,

$$\frac{\pi_{N,\sigma}(A^c_{N,f})}{\pi_{N,\sigma}(f)} \longrightarrow 0.$$

All functions tied with the target's energy remain in this set. By cloning and compression, its log cardinality is at most $h = F_L(d, B_L(d, C))$.

Corollary 3 (Strong-prior cumulative learning)**.** *At every fixed L, for every $N \geq C$ there is $\sigma_0 > 0$ such that all $0 < \sigma < \sigma_0$ satisfy Theorem 1 with $m = 2^d$, $\beta = 2^{-d}$, and $h = F_L(d, B_L(d, C))$. Consequently the total expected and high-probability mistake bounds are polynomial in d, C (and $\log(1/\delta)$ for high probability), independently of N. The threshold σ_0 may depend on N, d, L, f and the activation; it need not be uniform in width. One fixed sufficiently small variance works for the entire infinite stream.*

The same bounds hold for any limit of these finite-domain posterior laws as $\sigma \downarrow 0$. No selection of a unique energy minimizer is required. Ordinary function-MAP prediction also obeys the corresponding finite-class random-order bound: when $\beta < 1$, every function outside $A_{N,f}$ has smaller prior mass than the consistent target and cannot be MAP. Ties must be selected independently of the next input. Parameter MAP and a choice of an unattained minimum-energy parameter are different notions and are not identified with function MAP here.

4.3 Every fixed exponent $\gamma > 1$

Set $m = 2^d$, $A = A_L(d, C)$, and

$$\begin{aligned} K &= 4A + 8\log(Lm), \qquad a = a_K, \\ T &= \left\lceil \max\{C, [8(d+1)]^{1/(\gamma-1)}, (8/a)^{2/(\gamma-1)}\} \right\rceil, \\ h &= F_L(d, (L+3)K), \qquad J = AT^\gamma/2. \end{aligned}$$

At $N \geq T$, the relative exceptional mass is at most

$$L\exp\{-(K/8 - A/2)N^\gamma\} = L(Lm)^{-N^\gamma} \leq 1/m.$$

The corresponding finite set necessarily contains f, since otherwise the ratio would be at least one. At $C \leq N < T$, $-\log \pi(f) \leq AN^\gamma/2 \leq J$.

Corollary 4 (Above-mean-field cumulative learning)**.** *For every fixed L and $\gamma > 1$, Theorem 1 applies at $N \geq T$ with the preceding $h, \beta = 1/m$; Lemma 2 applies at $C \leq N < T$ with cost J. Taking the maximum of the two budgets gives total expected and high-probability mistake bounds polynomial in $d, C, \log(1/\delta)$, uniformly for every $N \geq C$. These are statements at the specified positive variances. The polynomial's degree may depend on L, γ; this proof makes no uniform claim as $\gamma \downarrow 1$.*

4.4 Mean field

For tanh at $\gamma = 1$, put

$$K = 4A + 8\log(Lm), \qquad T = T_L(d, K), \qquad h = G_L(d, K), \qquad J = AT/2.$$

The same calculation gives relative exceptional mass at most $1/m$ for $N \geq T$. Below T, the direct target cost is at most J.

Corollary 5 (Mean-field cumulative learning)**.** *The finite-variance mean-field tanh learner has total expected and high-probability mistake bounds polynomial in $d, C, \log(1/\delta)$, uniformly over every $N \geq C$, by combining Theorem 1 and Lemma 2 with the preceding budgets. The polynomial degree may depend on depth.*

Why 2^{-d} does not create an exponential sample cost. The relative tail is reduced to 2^{-d} by increasing the structural budget by $O(d)$. The log size of the finite relevant class is polynomial in this budget. One never requests an approximation accuracy of 2^{-d} from a compression theorem whose cost is polynomial in inverse accuracy.

4.5 An extension to ReLU

The ReLU manuscript [6] replaces bounded-activation compression by product-norm capacity control. Its independently reviewed structural bounds can be inserted into the same cumulative theorem. For a concrete depth-L, width-C ReLU margin teacher, set

$$P_* = \prod_{s=1}^{L} \|W_*^s\|_F, \qquad a_H = 1 + \sqrt{2(L-1)\log 2}, \qquad v_* = \lceil 4e(d+1)P_*^2 a_H^2 \rceil.$$

The ReLU strong-prior proof gives a finite relevant class with $h = v_* \log(m+1)$, independent of learner width. Choose the variance small enough at each fixed $N \geq C$ that its relative tail is at most $1/m$. Theorem 1 then gives expected total errors $O(v_* d^2)$, with the stated logarithmic-confidence extension. For a bounded-weight teacher, $P_*^2 \leq (d+1)C^{4L-2}$, so this is polynomial at fixed depth.

There is also a specified finite-variance result for ReLU, without a small-variance limit. The same manuscript proves the following energy tail and count. Define

$$\begin{aligned}
K_R &= L(4dP_*^2)^{1/L}, \qquad S_R = d + L, \\
r_R &= \min\{1, [4\sqrt{2L}(\sqrt{K_R/L} + 1)^{L-1}]^{-1}\}, \\
B_R &= \tfrac{1}{2}(\sqrt{K_R} + r_R)^2 + S_R \log\left(3\max\{1, \sqrt{S_R}/r_R\}\right), \\
T_R &= 4[B_R + (S_R/2)\log 2 + \log m], \\
v_R &= \left\lceil \frac{4(d+1)}{d}(T_R/L)^L a_H^2 \right\rceil.
\end{aligned}$$

For every $N \geq C$ and every $\gamma \geq 2$, the Gaussian with energy coordinate variance $N^{-\gamma}$, globally margin-conditioned as above, has relative function mass at most $1/m$ outside a finite class with $h = v_R \log(m+1)$. Thus Theorem 1 gives an all-width total mistake bound polynomial at fixed depth, uniformly over $\gamma \geq 2$. For one hidden layer, $L = 2$, the same statement holds for every $\gamma \geq 1$. The deeper finite-variance range $1 \leq \gamma < 2$ is not settled by this imported ReLU argument. These assertions use the ReLU paper's homogeneous balancing and capacity proofs, not an extension of the bounded-activation compression theorem by assertion.

5 Depth-polynomial support restrictions

In this section C denotes the width of a layered *Boolean* circuit whose stages fit the learner's hidden layers. Constant wires and any fixed-basis compilation overhead count toward its width and depth. This is the teacher convention of the neuron-merging results; it is a stronger requirement than a comparably small arbitrary real-valued neural teacher.

The same statistical proof applies to finite loss classes, not just Boolean hypotheses. Suppose a posterior of real-output networks admits, apart from posterior mass at most β at every history, a map into a finite collection of losses $\ell_h : \mathcal{X} \to [0,1]$ such that

$$\ell_h(X_i) = 0 \text{ on every training input}, \qquad \mathbf{1}\{f(x)z_w(x) \leq 1/4\} \leq \ell_h(x) \quad \text{for every } x.$$

For a loss with mean u, at least mu inputs have positive loss, so the probability that its empirical loss is zero on a size-n random prefix is at most e^{-nu}. Its remaining mean equals $m/(m-n)$ times its full-domain mean. Every step of Theorem 1 therefore applies unchanged. The map may depend on the retained network and the history; the finite loss class must be fixed before drawing the input order. Here the cumulative posterior error variable uses the tested margin event $fz_w \leq 1/4$; the resulting bounds also control sign errors.

The neuron-merging papers [4, 5] supply exactly this form of deterministic compression. Retained networks have a compressed grid representative whose clipped output differs uniformly by at most $3/32$. Training uses signed margin $> 1/2$, so the representative has signed clipped output $\geq 13/32$ on training inputs. A tested margin error $fz_w \leq 1/4$ gives representative signed clipped output at most $11/32$. Use a ramp loss equal to one at or below $11/32$ and zero at or above $13/32$. The resulting finite class has log size polynomial jointly in $d, L, C, c + c^{-1}, \log(1/\beta)$, independently of N and the scale $0 < t \leq 1$. Choosing $\beta = 1/m$ proves a total mistake bound polynomial in depth as well as width for those structural support restrictions. At fixed remaining parameters, the teacher-width dependence remains quadratic up to logarithms, with an extra factor polynomial in d.

This does not by itself establish a polynomial-depth total mistake bound under the weaker internal-noise restriction alone. Its current entropy compression couples predictions with an error probability α and has size polynomial in $1/\alpha$. Taking α of order 2^{-d} would destroy the desired dimension dependence. That is a limitation of this extension, not a counterexample to such a cumulative theorem. If the weaker support is additionally globally margin-conditioned, the exact finite-function mean-field argument still gives the fixed-depth cumulative result after using its robust target-mass cylinder; it does not preserve the improved polynomial depth dependence.

6 Which input assumptions matter?

Uniform random order is doing real work. A class of $m+1$ simple functions can require m mistakes in a prescribed order despite log cardinality only $\log(m+1)$. Let $f \equiv +1$ and let g_i differ from f only at point x_i. Give g_i prior weight R^{m-i+1} and f weight one, and reveal $x_1, \ldots, x_m$.

Before x_i, g_i has posterior mass greater than $1 - 1/R$, because the previous candidates have been eliminated. For $R > 3$, every round is a 2/3-confidence failure, majority prediction is wrong every time, and expected posterior-sampling mistakes exceed $(1 - 1/R)m$. All functions are individually simple point exceptions on the Boolean cube. This is a counterexample to an arbitrary-order conclusion from a finite relevant-class count alone. It is not a counterexample proved for the particular Gaussian neural-network priors.

An arbitrary iid input distribution is also different from uniform iid. In this same example, take probabilities proportional to $1, \rho, \ldots, \rho^{m-1}$. The chance that the first occurrence of x_{i+1} precedes that of x_i is $\rho/(1 + \rho)$. A union bound shows that the first-appearance order is the displayed adverse order with probability at least $1 - (m - 1)\rho$. Thus even iid sampling can be almost adverse when point probabilities range over sufficiently many scales. Again, this isolates a limitation of the statistical inference rather than proving failure of a specific neural-network prior.

A useful nonuniform extension. For any iid distribution on a finite support of size m and minimum positive probability $p_{\min}$, the same relevant-class assumptions give

$$\mathbb{E}S \le 3 + (h + 1)(1 + \log T) + m\beta, \qquad T = \left\lceil p_{\min}^{-1} \log(2/p_{\min}) \right\rceil .$$

Indeed, expected relevant-class risk after n labels is at most $(h + 1)/n$. It is also at most the expected unseen probability, which is at most $e^{-np_{\min}}$. Sum the first bound through T and the second afterwards. The outside-class contribution is at most β times the unseen probability at every history, and the sum of the expected unseen probabilities equals the support size m. In particular, the total bound stays polynomial whenever $\log(1/p_{\min})$ does. Uniform sampling is a clean special case, not the only useful one.

What this does not show about a converse. Few realized mistakes in one run do not alone imply that the target has a small circuit: favorable random guesses and incomplete domain coverage must be excluded. Nor does a general bounded-mistake learner automatically supply a small circuit unless its predictions have controlled computational descriptions. The results here establish the upper direction for specified Bayesian learners; a converse requires an additional argument.

A Explicit imported neural-network budgets

For reproducibility, the quantities below are the scalar bounds supplied by the earlier proofs. Their mathematical roles are stated in Section 4. They are intentionally loose.

A common norm-sublevel count. For $J \ge 1$, put

$$\begin{aligned} \rho_J &= \frac{2^{-L}(J + 1)^{-(L-1)/2}}{12(L - 1)}, \\ u &= \max\left\{ \lceil 9(J + 1) \rceil, \left\lceil 256\rho_J^{-4}[d + \log(L/\rho_J)] \right\rceil \right\}, \\ F_L(d, J) &= [u(d + 2) + (L - 2)u^2] \log\left(3 + 16Lu \max\{u, d + 1\}[u(u + 1)]^{L-1}\right). \end{aligned}$$

The norm-compression theorem preserves all Boolean signs and yields a width-u, weight-bounded implementation. Uniform parameter rounding gives the count F_L. The class is common to all original learner widths.

The target cylinder. Set

$$\begin{aligned} b_L &= \sqrt{16\log 9 + 8\log(4L)}, \qquad a_0 = 2 + 3C^2 + b_L, \\ \rho_0 &= [2 + 9C^2(d + L + 1)a_0^{L-2}]^{-1}, \\ B_0 &= d(d+1)C^2 + 2(L-2)C^4 + 36C^4, \\ A_L(d, C) &= 2B_0 + 4[d + 2 + (L-2)C]/\rho_0 + 2. \end{aligned}$$

The earlier cylinder proof gives raw target mass at least $\exp(-A_L(d, C)N^\gamma/2)$ for every $\gamma \geq 1, N \geq C$.

Mean-field exact-function compression. Let $r = L - 1$, $K \geq 4$, and choose backwards

$$g_r = 32rK^2, \qquad g_s = 8g_{s+1}(1 + g_{s+1})^2 \quad (s = r-1, \ldots, 1).$$

Put

$$q = \lceil g_1^2/2 \rceil, \quad \ell = \log(2 \cdot 2^{2d} + 1), \quad T_L(d, K) = \lceil \max\{1, 2q\ell/(K-1)\} \rceil .$$

For the finite-description bound, write $g = g_1$, $b = 8(1+g)^2$, and set

$$\begin{aligned} \eta &= [32Krb^{r-1}]^{-1}, & u_* &= \lceil 2\eta^{-2}\log(8r\,2^{2d}) \rceil, \\ D_* &= u_*(d+1) + (L-2)u_*^2, & B_* &= \max\{1, \sqrt{4g + 2\log(8\max\{D_*, 1\})}, 4K/u_*\}, \\ P_* &= D_* + u_*, & \Lambda_* &= (d + Lu_*)[1 + u_*(B_* + 1)]^{L-1}. \end{aligned}$$

Then $G_L(d, K) = P_* \log(3 + 16B_*\Lambda_*)$ is valid. At fixed depth, T_L and G_L have polynomial majorants in d, K. The covariance compression uses the global margin to recover exactly the whole Boolean sign function. It is not merely an average-error approximation.

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